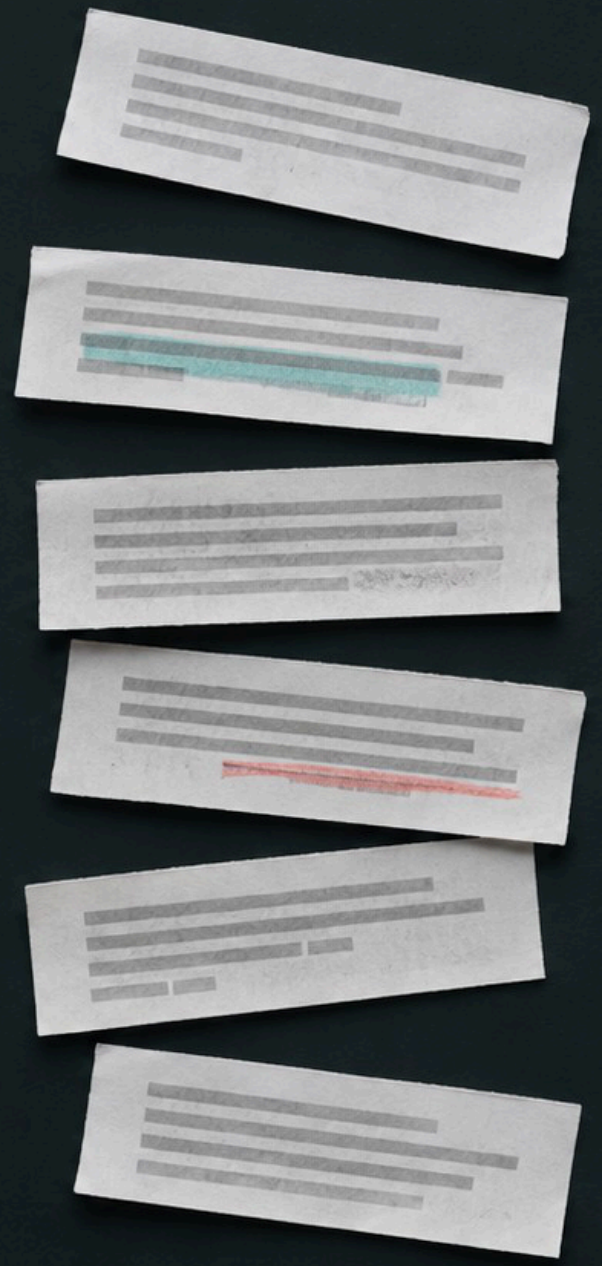
DAMDID 2026

# Deep Re-annotation and Controlled Candidate Pools

Re-evaluating ranking models on MS MARCO



# A relevant passage can receive zero gain

## “what causes boiling point to increase”

| Passage (shortened paraphrase) | Original qrels | Human judgment |
|---|---|---|
| 7112148 Adding salt raises the boiling point. | Relevant | Relevant |
| **7399382 A non-volatile solute raises the boiling point.** | Unjudged | **Relevant** |
| 987600 The boiling temperature of acetone. | Unjudged | Not relevant |

qrels: query relevance judgments. Unjudged passages receive zero gain.

# Data and judgment coverage

English MS MARCO

## 8.84 million passages

### Human judgments

**1,179**

queries; 33,667 pairs

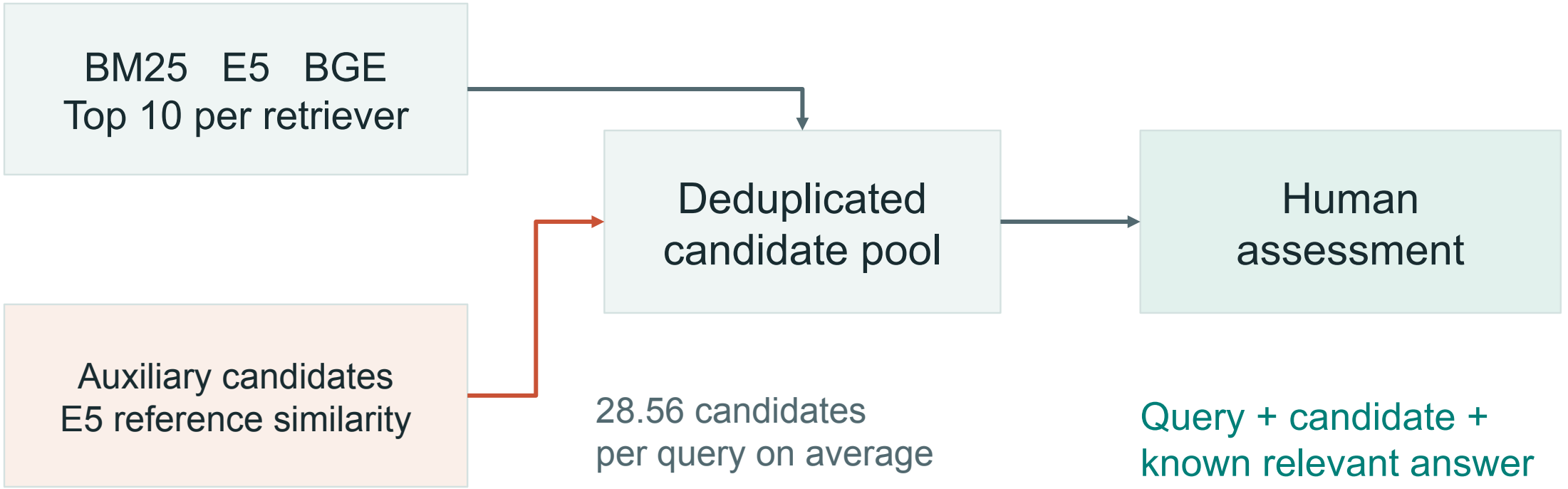
### Judge predictions

**101,093**

queries; varying pool depths

“Deep” describes judgment coverage within selected candidate pools.

# Reference-assisted manual annotation




Reference information supports candidate preparation and human assessment.

# More relevant judgments within the same pools

Mean passages judged relevant per query

**1.05**

**17.47**

## Original qrels

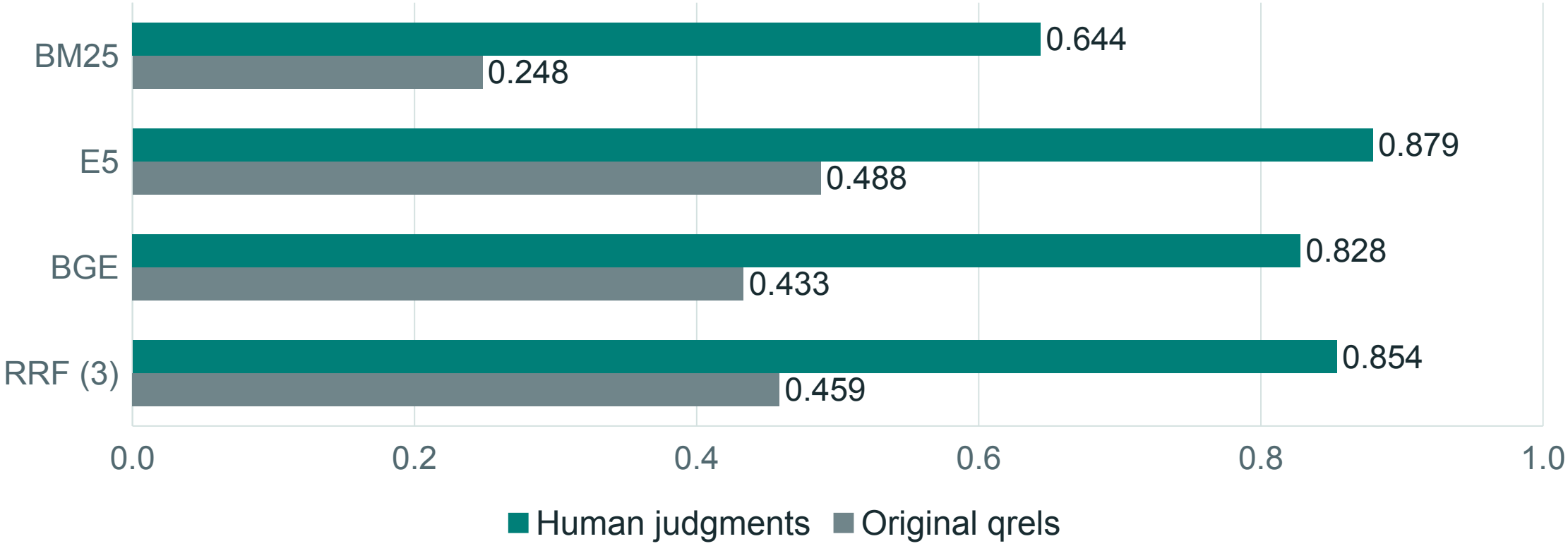
1,243 positive pairs

## Extended human judgments

20,597 positive pairs

1,179 queries. Same candidate pools and retrieval rankings.

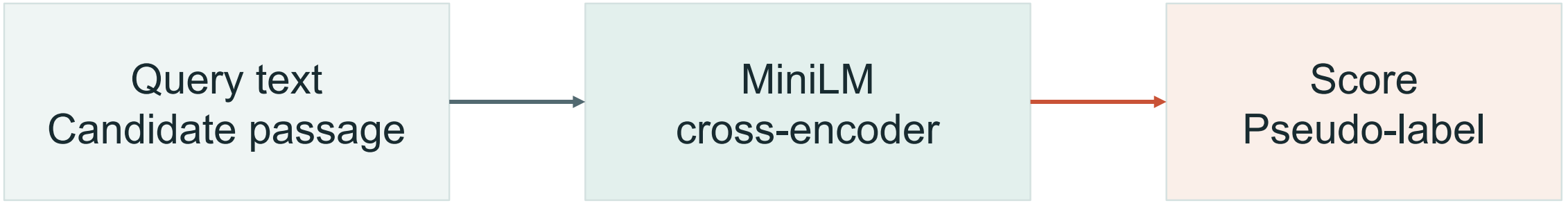
# Denser judgments change measured NDCG




**E5 paired change: +0.391  95% CI [0.365, 0.417]**

Pooled NDCG@10. Same rankings; shared ideal denominator within each query and label set.

# A compact judge extends label coverage

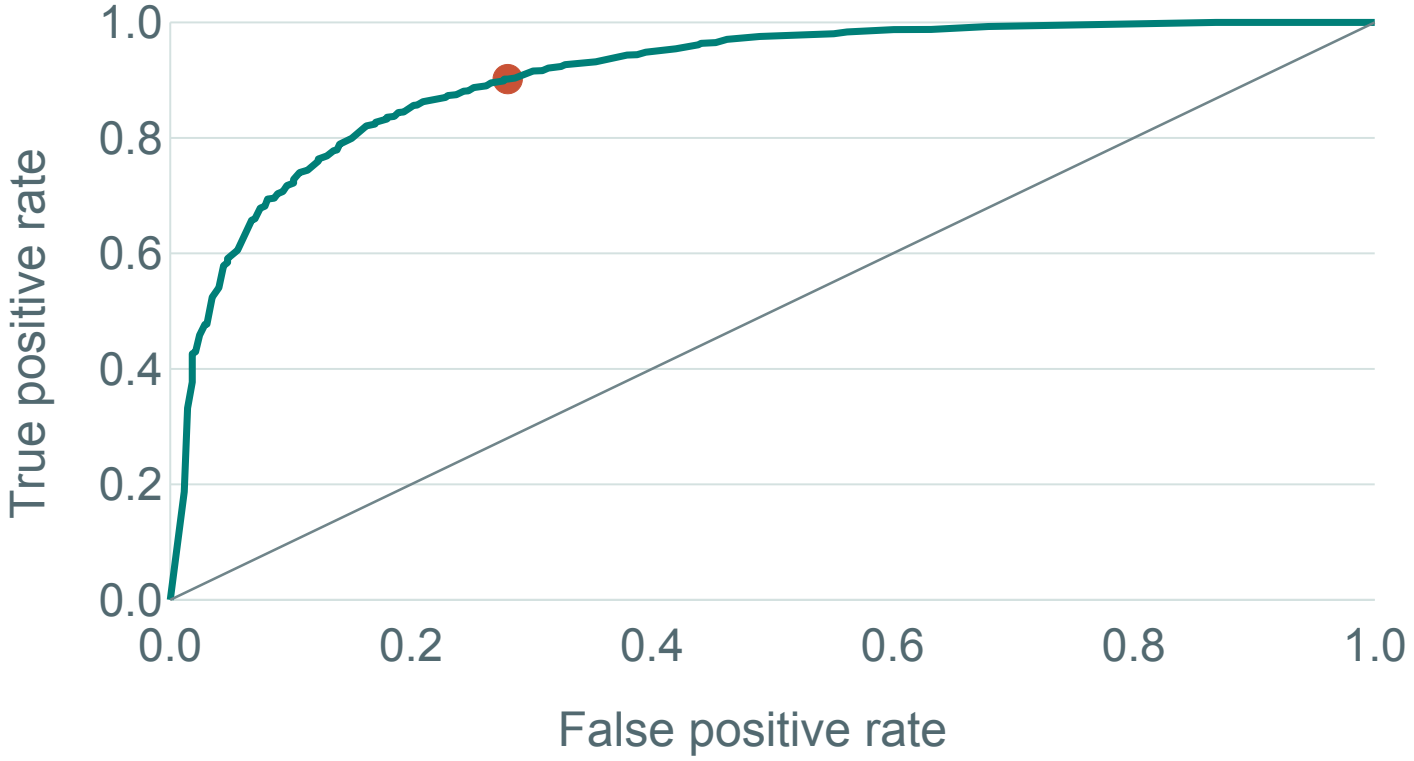



Jointly scores the two texts

Inputs exclude the reference answer, ranks and retrieval scores.

Pseudo-labels are model predictions.

# Judge agreement on held-out queries




**0.907**

ROC-AUC

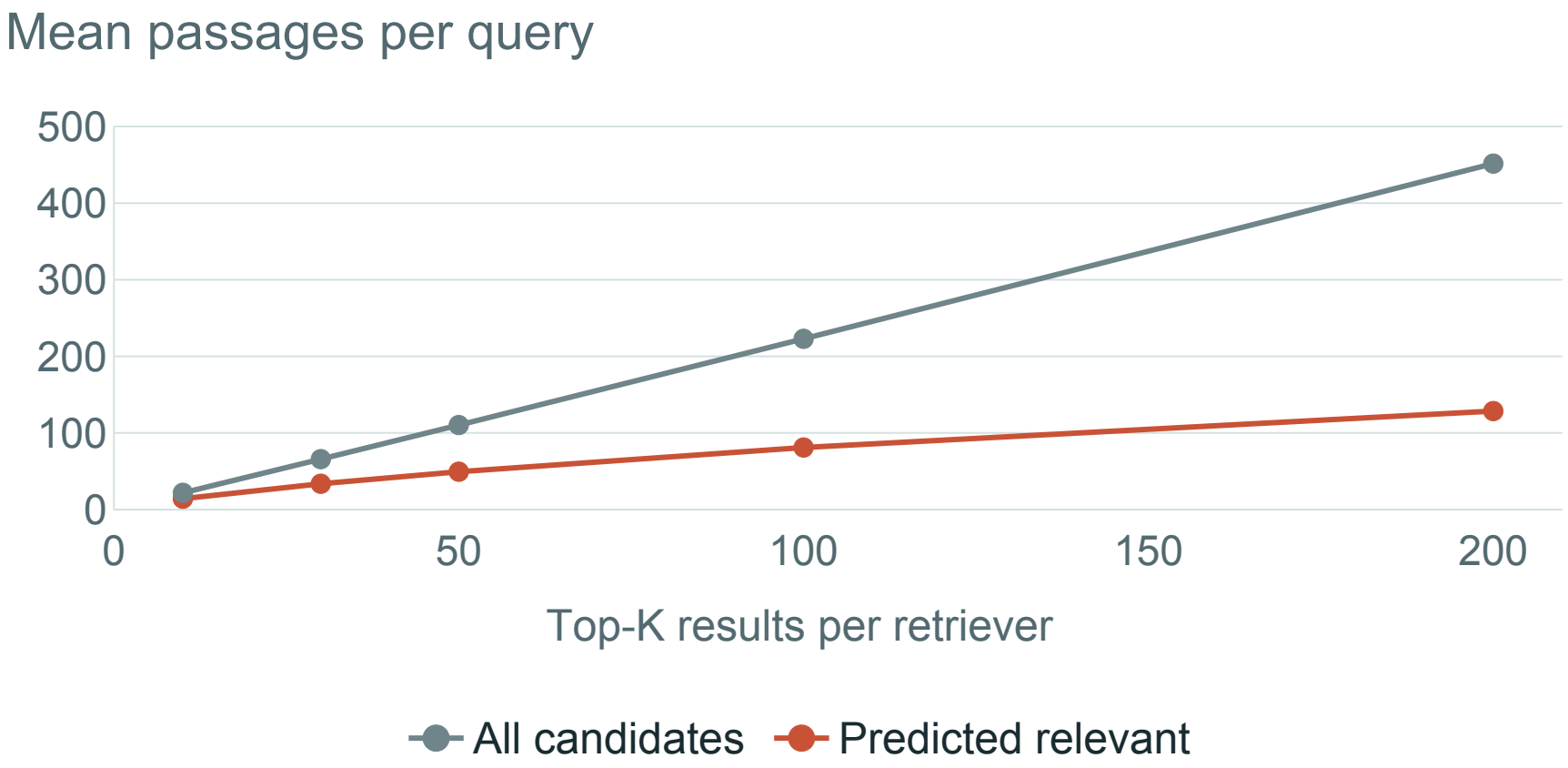
**0.875**

F1 at validation threshold

Marked point: threshold selected on validation queries

177 held-out manual-pool queries; 5,002 pairs. Deeper pools lack human validation.

# Larger pools contain more predicted positives




**451.5**

candidates

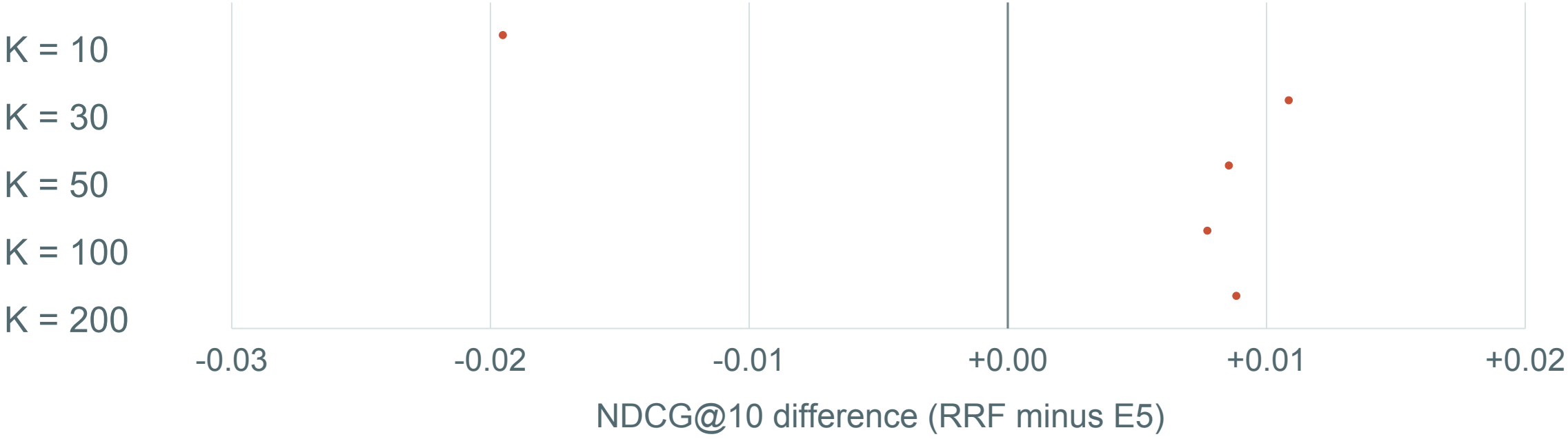
**128.6**

predicted relevant at K = 200

101,093 queries, including zero-positive pools. Same judge and threshold.

# RRF and E5 across pool depths

RRF(L = K) vs E5. Pool and fusion input depths vary together.

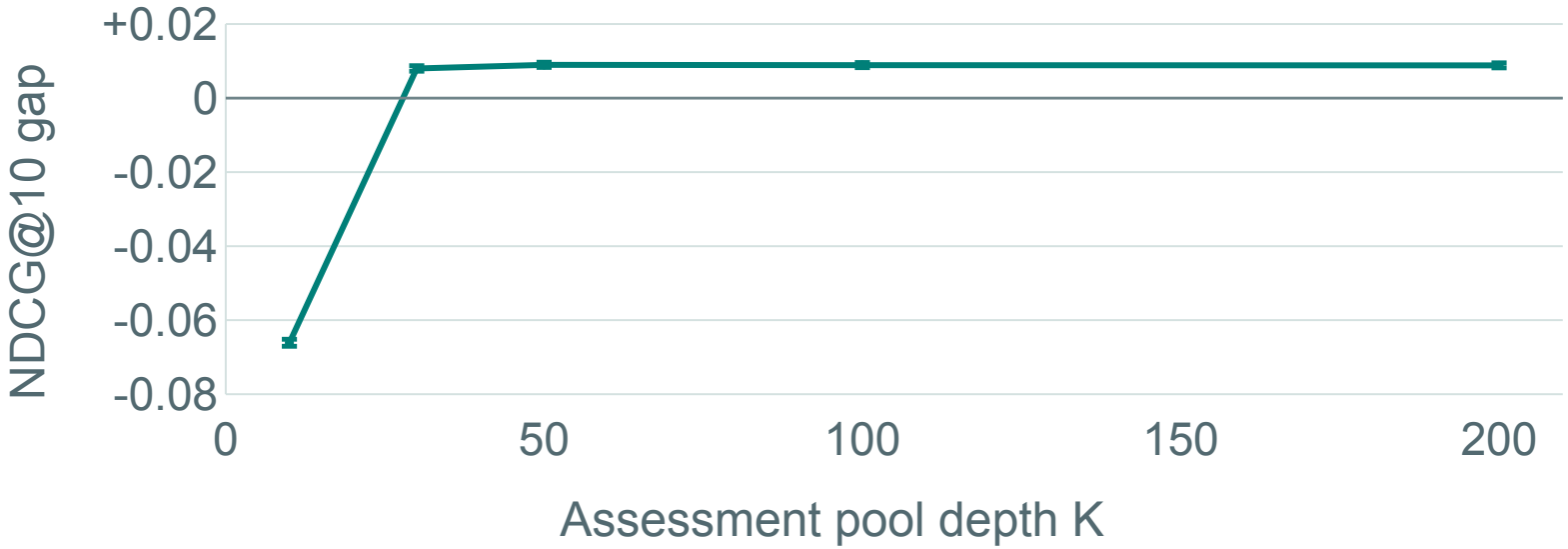



**K = 200: +0.0088  95% CI [0.0082, 0.0095]**

Single-judge pseudo-labels. Mean differences with 95% paired query-bootstrap intervals.

# Pool coverage can reverse a comparison

**Fixed RRF inputs: top 200. Both rankings remain unchanged.**




**14.4%**

of RRF top-10 pairs outside the pool at K = 10

145,643 query–passage pairs

**K = 10: −0.0661**

**K = 30: +0.0080**

RRF minus E5 under pseudo-labels. Unjudged returned passages receive zero gain.

# Scope and robustness

## Sensitivity checks

| Evaluation | Remaining queries | Observed result |
|---|---|---|
| Manual labels<br>23 affected queries excluded | 1,156 | System order unchanged |
| Full-set pseudo-labels<br>Manual queries excluded | 99,914 | Signs of RRF–E5 differences unchanged |

Inter-assessor agreement is unmeasured.
Full-set results lack independent human validation.

Evidence concerns selected English MS MARCO pools and one judge.

# Judgment coverage affects retrieval evaluation

## Human judgments

## E5 NDCG@10: 0.488 to 0.879

Same ranking; more complete relevance judgments

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## Fixed rankings; single-judge pseudo-labels

The judgment pool can change which system scores higher

Evidence from selected English MS MARCO candidate pools.

# NDCG@10 and paired annotation effects

## NDCG@10 = DCG@10 / ideal DCG@10

| Ranking | Human − original | 95% paired CI |
|---|---|---|
| BM25 | +0.396 | 0.372 to 0.419 |
| E5 | +0.391 | 0.365 to 0.417 |
| BGE | +0.395 | 0.369 to 0.419 |
| RRF (3) | +0.395 | 0.371 to 0.419 |

5,000 paired query bootstraps. All four annotation effects: $p < 0.001$.

# Judge test protocol and observed errors

Train queries

**825**

Validation queries

**177**

Test queries

**177**

| Test target | Predicted negative | Predicted positive |
|---|---|---|
| Coded negative | 1,300 | 506 |
| Coded positive | 314 | 2,882 |

**Precision 0.851** **Recall 0.902** **F1 0.875**

5,002 pairs; six unresolved targets mapped to zero and one empty-text record.

# Full-set depth results

| K | Candidates / query | Predicted relevant / query | E5 NDCG@10 | RRF L=K NDCG@10 | RRF L=200 NDCG@10 |
|---|---|---|---|---|---|
| 10 | 21.9 | 14.2 | 0.871 | 0.852 | 0.805 |
| 30 | 65.8 | 33.7 | 0.851 | 0.862 | 0.859 |
| 50 | 110.4 | 49.4 | 0.847 | 0.856 | 0.856 |
| 100 | 222.9 | 81.0 | 0.844 | 0.851 | 0.853 |
| 200 | 451.5 | 128.6 | 0.842 | 0.850 | 0.850 |

Single-judge pseudo-labels. All 101,093 queries remain in every average.

# Sensitivity to query exclusions

Human-label NDCG@10

| Ranking | All 1,179 queries | 1,156 after exclusions |
|---|---|---|
| BM25 | 0.644 | 0.644 |
| E5 | 0.879 | 0.879 |
| BGE | 0.828 | 0.828 |
| RRF (3) | 0.854 | 0.855 |

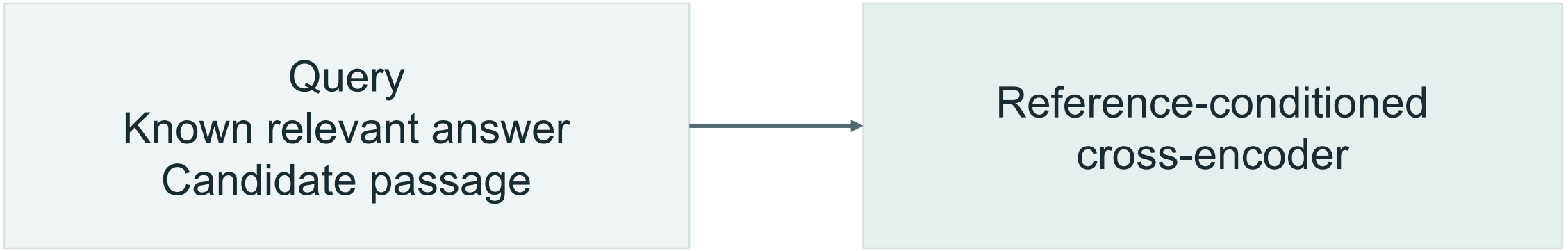
**Full set without manual queries: N = 99,914**

At K = 200: RRF − E5 = +0.00877

Exclusion checks do not replace independent human assessment.

# Next validation: reference-conditioned judging

Proposed input




**Compare stronger judges against independent human assessments.**

Planned evaluation. No results for this model are reported in the present study.